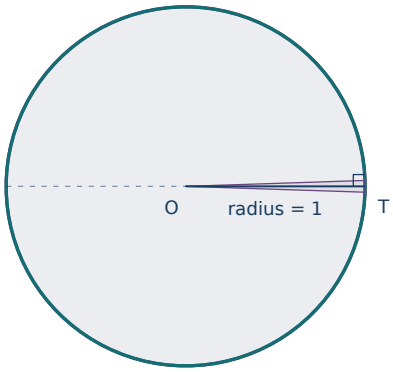
STOICHEIA / ARCHIMEDES, CIRCLE MEASUREMENT 3

Keep the safe side.

A saved experiment at 96 sides. One altered approximation: outer path, row 2, Y = 11823/20. This trial is kept separate from the printed numbers.

Open this experiment · Save editable JSON · Save the exact ledger

96 sides · equal horizontal and vertical scales



The gaps are now too small to judge by eye.

Proposed endpoints · proof chain broken



The scale expands as the interval narrows.

Inner 96-gon · certified lower bound		Outer 96-gon · unproved candidate
25344/8069	< c/d ?	587520/186941
≈ 3.140909654		≈ 3.142809764

The diagram is unchanged by arithmetic trials. An unproved candidate is not asserted to bound the circumference. A rejected proof does not establish that its final numerical answer is false.

Your two proof paths

Outer path: keep A below the true ratio, so n/A bounds C/d from above.

Sides	A after adding	H after rounding	Local square test	Length estimates	n / A
6	265/153	2	safe	valid	918/265 upper bound
12 *	571/153	3941/1020	wrong side	broken	1836/571 upper bound
24	23243/3060	9377/1224	safe	broken	73440/23243 candidate only
48	93371/6120	3119/204	safe	broken	293760/93371 candidate only
96	186941/6120	not needed	addition only	broken	587520/186941 candidate only

Inner path: keep H above the true ratio, so n/H bounds C/d from below.

Sides	A after adding	H after rounding	Local square test	Length estimates	n / H
6	1351/780	2	safe	valid	3 lower bound
12	2911/780	2411/624	safe	valid	7488/2411 lower bound
24	1823/240	20227/2640	safe	valid	63360/20227 lower bound
48	1007/66	6055/396	safe	valid	19008/6055 lower bound
96	12097/396	8069/264	safe	valid	25344/8069 lower bound

The altered step

In the printed scale, the chosen hypotenuse is **Y = 11823/20**; the other scale is **153**.

$$Y^2 \le 349450$$

This is the required local test. Its exact signed remainder is

`349450 − Y² =` **`-3329/400.`**

The chosen approximation is on the wrong side: this step does not justify its new length estimate. Later arithmetic cannot restore the missing implication.

The printed choices, without your alteration

Outer path: keep A below the true ratio, so n/A bounds C/d from above.

Sides	A after adding	H after rounding	Local square test	Length estimates	n / A
6	265/153	2	safe	valid	918/265 upper bound
12	571/153	4729/1224	safe	valid	1836/571 upper bound
24	1033/136	9377/1224	safe	valid	3264/1033 upper bound
48	9337/612	3119/204	safe	valid	29376/9337 upper bound
96	9347/306	not needed	addition only	valid	29376/9347 upper bound

Inner path: keep H above the true ratio, so n/H bounds C/d from below.

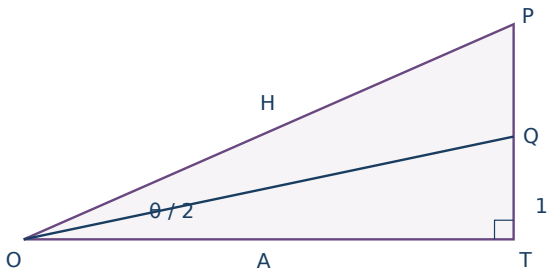
Sides	A after adding	H after rounding	Local square test	Length estimates	n / H
6	1351/780	2	safe	valid	3 lower bound
12	2911/780	2411/624	safe	valid	7488/2411 lower bound
24	1823/240	20227/2640	safe	valid	63360/20227 lower bound
48	1007/66	6055/396	safe	valid	19008/6055 lower bound
96	12097/396	8069/264	safe	valid	25344/8069 lower bound

The source-number 96-gons establish **25344/8069 < C/d < 29376/9347**. These fit strictly within **223/71 < C/d < 22/7**. They are bounds, not evaluations of the circle ratio.

Printed square-root choices, before normalization

Path	Sides	Y	Scale	Printed page
outer	6	265	153	264
outer	12	591 + 1/8	153	264
outer	24	1172 + 1/8	153	264
outer	48	2339 + 1/4	153	266
inner	6	1351	780	266
inner	12	3013 + 3/4	780	268
inner	24	1838 + 9/11	240	268
inner	48	1009 + 1/6	66	270
inner	96	2017 + 1/4	66	270

Why an approximation has a direction



A schematic right triangle, with the angle at O bisected at Q. The drawing illustrates the relation; it is not the selected polygon at its current scale.

Normalize the outer half-side TP to 1. The radius OT becomes A; the slant OP becomes H. Thus **H² = A² + 1**.

The angle-bisector theorem gives PQ / QT = H / A, and PQ + QT = 1. Therefore QT = A / (A + H). Divide the smaller triangle's lengths by QT: its adjacent leg becomes **A′ = A + H**, and its hypotenuse becomes **H′ = √(A′² + 1)**.

Halving the angle doubles the side count. The outer polygon's perimeter / diameter is **n / A**. A similar right triangle subtended by an inner chord gives diameter / chord

= H, so the inner polygon's perimeter / diameter is **n / H**.

1. **Outer path:** start below √3. Adding lower estimates stays below the exact sum; choose each new H no larger than √(A² + 1). Inverting A makes n/A an upper estimate of the outer perimeter ratio, which already exceeds C/d.
2. **Inner path:** start above √3. Add upper estimates and choose H no smaller than √(A² + 1). Inverting H makes n/H a lower estimate of the inner perimeter ratio, which already falls below C/d.
3. **Compare squares:** all lengths are positive, so Y ≤ √R exactly when Y² ≤ R. No numerical square root is needed to decide. A safe step after an unsafe step cannot restore the proof chain.

This is a modern unified notation for the two geometric arguments. The printed inner calculation rescales its ratios from 780 to 240, then to 66; exact fractions preserve those equalities. The six-sided seeds use H = 2 exactly. The outer 96-gon needs only A, so no final H is invented.

Eight readings, with sixteen word notes

Selections from J. L. Heiberg's 1880 volume I, pp.262–271. Greek is the ancient text in a later critical edition; Latin is Heiberg's translation, not an ancient parallel. English and teaching notes are new. Line breaks, hyphenation and geometric-letter shapes are normalized; inline reference markers are omitted. Full pages retain apparatus, brackets and adjacent proposition 2. That proposition's 11:14 area ratio must not be imported into proposition 3 as an exact equality.

1. The circumference and the diameter

Παντὸς κύκλου ἡ περίμετρος τῆς διαμέτρου τριπλασίων ἐστί

Cuiusuis sphaerae perimetrus diametro triplo maior est

For every circle, the circumference is three times the diameter …

The sentence continues with an excess: it does not assert C = 3d. Heiberg's Latin actually prints sphaerae, ‘of a sphere’, where the Greek has κύκλου, ‘of a circle’. We retain that discrepancy. Read this English from the Greek; the diagrams concern a plane circle.

περίμετρος / perimetrus / ambitus

the boundary's length; for this circle, circumference

τῆς διαμέτρου / diametro

‘of the diameter’ in the Greek comparison; the whole way across, not the radius

Greek p.262; Latin p.263.

---

2. Two directions of excess

καὶ ἔτι ὑπερέχει ἐλάσσονι μὲν ἢ ἑβδόμῳ μέρει τῆς διαμέτρου, μείζονι δὲ ἢ δέκα ἑβδομηκοστομόνοις.

et praeterea excedit spatio minore, quam septima pars diametri est, maiore autem quam 10/71.

… and exceeds that by less than a seventh part of the diameter, but by more than ten seventy-first parts.

The paired μὲν … δὲ keeps two different comparisons together. Adding three to each excess gives 223/71 < C/d < 22/7. The Latin's stacked fraction is written 10/71 here. Neither endpoint is an equality.

ἐλάσσονι μὲν … μείζονι δὲ / minore … maiore autem

less on the one hand … greater on the other; two sides of a bound

ἑβδόμῳ μέρει / septima pars

a seventh part; ‘seventh’ is an ordinal used for a fractional part

Greek p.262; Latin p.263.

---

3. The line that touches

ἔστω κύκλος, καὶ διάμετρος ἡ ΑΓ, καὶ κέντρον τὸ Ε, καὶ ἡ ΓΛΖ ἐφαπτομένη

sit circulus, et diametrus ΑΓ, et centrum Ε, et ΓΛΖ linea circulum contingens.

Let there be a circle, with diameter ΑΓ, centre Ε, and tangent ΓΛΖ.

The radius at the point of contact is perpendicular to the tangent. The outer half-side sits on that tangent. Our diagrams rotate the construction and use O, T and P so the modern arithmetic does not masquerade as a facsimile.

ἔστω / sit

let there be; an instruction to establish the figure

ἐφαπτομένη / contingens

touching; the feminine participle supplies ‘tangent [line]’

Greek p.264; Latin p.265.

---

4. Halve the angle

τετμήσθω οὖν ἡ ὑπὸ ΖΕΓ δίχα τῇ ΕΗ.

iam secetur ∠ ΖΕΓ in duas partes aequales linea ΕΗ.

Therefore let angle ΖΕΓ be cut in two by ΕΗ.

δίχα says ‘in two’. The argument uses equal halves: Heiberg's Latin makes aequales explicit. Halving the angle doubles the polygon's side count; it does not halve the circumference. Four bisections take six sides to ninety-six.

τετμήσθω / secetur

let it have been cut; a geometrical command

δίχα / in duas partes aequales

in two; here two equal angular parts

Greek p.264; Latin p.265.

---

5. Greater is useful

ὥστε ἡ ΓΕ πρὸς ΓΗ μείζονα λόγον ἔχει, ἤπερ φοα′ πρὸς ρνγ′.

quare ΓΕ : ΓΗ > 571 : 153.

Thus ΓΕ has to ΓΗ a greater ratio than 571 has to 153.

φοα′ = 500 + 70 + 1; ρνγ′ = 100 + 50 + 3. These Greek letters are numerals, distinct from the diagram's point labels. A lower bound on radius / half-side gives an upper bound on perimeter / diameter when the positive ratio is inverted.

μείζονα λόγον ἔχει / maiorem rationem habet

has a greater ratio; a strict comparison, not an approximate equality

πρὸς / ad / :

to, in comparing one magnitude with another

Greek p.264; Latin p.265.

---

6. The circle stays inside

ἡ τοῦ κύκλου ἄρα περίμετρος πολὺ μᾶλλον ἐλάσσων ἐστίν, ἢ τριπλασίων καὶ ἑβδόμῳ μέρει μείζων.

quare ambitus circuli multo magis minor est quam triplo et septima parte maior diametro.

The circle's circumference, then, is all the more less than three and a seventh times the diameter.

The circumscribed polygon already falls below the claimed upper limit. The circle's circumference is smaller still. πολὺ μᾶλλον strengthens the implication; it is not a numerical estimate of the gap.

πολὺ μᾶλλον / multo magis

all the more; the comparison passes from outer polygon to circle

ἐλάσσων / minor

smaller; keep its direction when following the nested perimeters

Greek p.266; Latin p.267.

---

7. Invert, and change direction

ἀνάπαλιν ἄρα ἡ περίμετρος τοῦ πολυγώνου πρὸς τὴν διάμετρον μείζονα λόγον ἔχει

perimetrus polygoni ad diametrum maiorem rationem habet quam 6336 : 2017¼

Inversely, therefore, the polygon's perimeter has to the diameter a greater ratio …

These selections align the resulting comparison rather than identical sentence boundaries. The Latin's preceding bracketed passage supplies the inversion and the 96 sides. Our 2017¼ writes the printed mixed fraction on one line. The inner construction bounds diameter / chord from above; inversion supplies a lower perimeter bound.

ἀνάπαλιν / e contrario [in the preceding passage]

inversely; a positive reciprocal reverses an inequality

τοῦ πολυγώνου / polygoni

of the polygon; at this step it is inscribed in the circle

Greek p.270; Latin p.271.

---

8. A conclusion with an excess

ἡ ἄρα τοῦ κύκλου περίμετρος τῆς διαμέτρου τριπλασίων ἐστί

itaque ambitus circuli triplo maior est diametro

Therefore the circle's circumference is three times the diameter …

The following words again retain an excess strictly between 10/71 and 1/7. This excerpt ends before them, just as the opening excerpt did. Read both halves together. The conclusion supplies bounds, not a final value π = 22/7.

ἄρα / itaque

therefore; marks the conclusion carried by the preceding argument

τριπλασίων / triplo

threefold; the continuing excess prevents reading this as exact equality

Greek p.270; Latin p.271.

Read the edition as well as the numbers

Pages 264–265 print equality at some approximation steps: in particular 265:153 in place of √3, and 591⅛:153 in place of a hypotenuse ratio. Those cannot be exact equalities: $265^2 = 70225$ whereas $3 \times 153^2 = 70227$. The workshop explicitly supplies and verifies sufficient directed inequalities. It does not claim these are the signs printed at every step.

Page 263's Latin statement prints *sphaerae*, ‘of a sphere’, for Greek κύκλου, ‘of a circle’. The reader retains both. Adjacent proposition 2 prints the area ratio 11:14; this workshop does not treat it as an exact circle identity. Heiberg's apparatus and notes make the transmitted calculation's editorial difficulties visible. The numbers are an edition's witness, not an autograph of Archimedes.

What the experiment establishes

All arithmetic decisions use reduced integer fractions. Decimal labels and polygon drawings use floating point only for display. The polygon perimeter inequalities are geometric premises of this model; this page does not prove every result about arc length, reconstruct ancient computational tools, or identify how Archimedes discovered the square-root bounds. The custom approximation is your modern intervention. One altered choice is retained at a time.

This file contains the two paths, settings, source readings, ten complete pages, font and licenses. It runs no scripts and requests no remote subresources. External links require a connection. Print to A4 for a reading copy; the circle is an equal-scale illustration, not a physical calibration template.

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Source edition

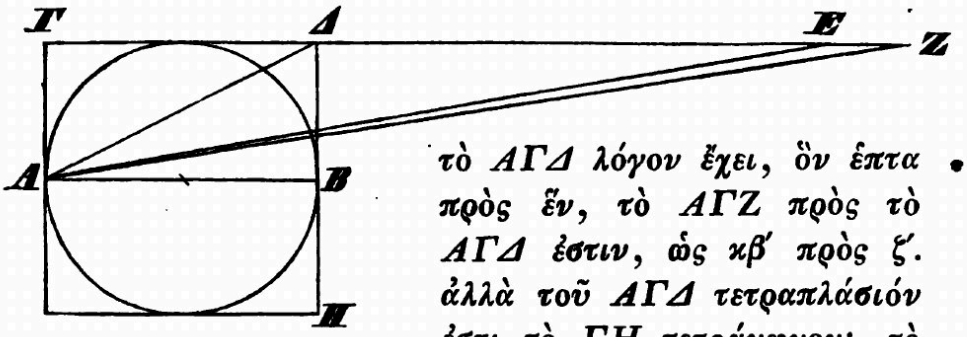
Archimedes, *Archimedis opera omnia cum commentariis Eutocii*, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1880, pp.262–271. Complete digitized volume. PDF page = printed page + 19. All retained pages are complete reductions, without retouching.

Heiberg 1880 / printed p.262 / Greek / PDF page 281

β΄.

Ὁ κύκλος πρὸς τὸ ἀπὸ τῆς διαμέτρου τετράγωνον λόγον ἔχει, ὃν ια΄ πρὸς ιδ΄.

ἔστω κύκλος, οὗ διάμετρος ἡ ΑΒ, καὶ περιγεγράφθω τετράγωνον τὸ ΓΗ, καὶ τῆς ΓΔ διπλῆ ἡ ΔΕ, ἕβδομον δὲ ἡ ΕΖ τῆς ΓΔ. ἐπεὶ οὖν τὸ ΑΓΕ πρὸς τὸ ΑΓΔ λόγον ἔχει, ὃν κα΄ πρὸς ζ΄, προς δὲ τὸ ΑΕΖ



τὸ ΑΓΔ λόγον ἔχει, ὃν ἑπτα πρὸς ἕν, τὸ ΑΓΖ πρὸς τὸ ΑΓΔ ἐστιν, ὡς κβ΄ πρὸς ζ΄. ἀλλὰ τοῦ ΑΓΔ τετραπλάσιόν ἐστι τὸ ΓΗ τετράγωνον· τὸ δὲ ΑΓΔΖ τρίγωνον τῷ ΑΒ κύκλῳ ἴσον ἐστίν [ἐπεὶ ἡ μὲν ΑΓ κάθετος ἴση ἐστὶ τῇ ἐκ τοῦ κέντρου, ἡ δὲ βάσις τῆς διαμέτρου τριπλασίων καὶ τῷ ζ″ ἔγγιστα ὑπερέχουσα δειχθήσεται]. ὁ κύκλος οὖν πρὸς τὸ ΓΗ τετράγωνον λόγον ἔχει, ὃν ια΄ πρὸς ιδ΄.

γ΄.

Παντὸς κύκλου ἡ περίμετρος τῆς διαμέτρου τριπλασίων ἐστί, καὶ ἔτι ὑπερέχει ἐλάσσονι μὲν ἢ ἑβδόμῳ μέρει τῆς διαμέτρου, μείζονι δὲ ἢ δέκα ἑβδομηκοστομόνοις.

---

1. β΄] om. F. 3. ιδ΄ ἔγγιστα Wallis. numeros lineolis transuersis supra ductis notat F. 5. διπλῆ] διπλασία Nizze. 9. ΑΓΖ ἄρα Wallis. 12. Post τετράγωνον Wallis addit: τὸ ἄρα ΑΓΖ τρίγωνον πρὸς τὸ ΓΗ τετράγωνον λόγον ἔχει, ὃν κβ΄ πρὸς κη΄, ἢ ὃν ια΄ πρὸς ιδ΄. 13. ΑΓΔΖ] sic F, Cr.;

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Heiberg 1880 / printed p.263 / Latin / PDF page 282

II.

Circulus ad diametrum quadratam eam rationem habet, quam 11 : 14.

sit circulus, cuius diametrus sit AB, et circumscribatur quadratum ΓH, et sit $\Delta E = 2\Gamma\Delta$, et $EZ = \frac{1}{7}\Gamma\Delta$. iam quoniam est $A\Gamma E : A\Gamma\Delta = 21 : 7$ [Eucl. VI, 1], sed $A\Gamma\Delta : AEZ = 7 : 1$ [Eucl. VI, 1], erit

$$A\Gamma Z : A\Gamma\Delta = 22 : 7.^{1})$$

sed $\Gamma H = 4A\Gamma\Delta$ [Eucl. I, 34], et triangulum $A\Gamma\Delta Z$ circulo AB aequale est [quia altitudo $A\Gamma$ radio aequalis est, basis autem triplo et praeterea septima parte maior diametro, hoc est ambitui proxime aequalis, ut demonstrabitur prop. 3; tum u. prop. 1].[2]) quare circulus ad quadratum ΓH eam rationem habet, quam 11 : 14.[3])

III.

Cuiusuis sphaerae perimetrus diametro triplo maior est, et praeterea excedit spatio minore, quam septima pars diametri est, maiore autem quam $\frac{10}{71}$.

---

1) Nam *ἀνάπαλιν* (Eucl. V, 7 *πόρ.*) $AEZ : A\Gamma\Delta = 1 : 7$; tum addendo sequitur proportio. sed poterat statim concludi ex Eucl. VI, 1; nam $\Gamma Z = (3 + \frac{1}{7})\ \Gamma\Delta = \frac{22}{7}\ \Gamma\Delta$.

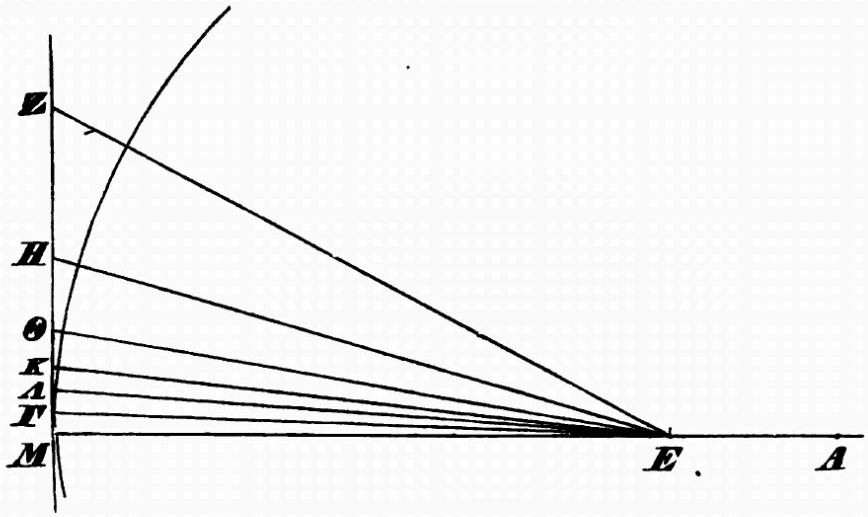
2) Hic locus *ἐπεί* lin. 13 — *δειχθήσεται* lin. 16 mire corruptus et confusus transscriptori tribuo, qui eum addidit, postquam prop. 2 et 3 permutauit; neque enim Archimedes hanc propositionem ante prop. 3, quo nititur, posuit.

3) Citatur haec propositio a Pseudoherone Geom. 103 p. 186.

$A\Gamma Z$ ed. Basil., uulgo. 15. Post *βάσις* Wallis addit: *τῆ τοῦ κύκλου περιμέτρῳ, ἥτις.* *τῷ*] scripsi; *τον* F, uulgo. 17. *ιδ' ἔγγιστα* Wallis.

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ἔστω κύκλος, καὶ διάμετρος ἡ ΑΓ, καὶ κέντρον τὸ Ε, καὶ ἡ ΓΔΖ ἐφαπτομένη, καὶ ἡ ὑπὸ ΖΕΓ τρίτον ὀρθῆς. ἡ ΕΖ ἄρα πρὸς ΖΓ λόγον ἔχει, ὃν τς′ πρὸς ρνγ′. ἡ δὲ ΕΓ πρὸς τὴν ΓΖ λόγον ἔχει, ὃν σξέ πρὸς ρνγ′. τετμήσθω οὖν ἡ ὑπὸ ΖΕΓ δίχα τῇ ΕΗ. ἔστιν ἄρα, ὡς ἡ ΖΕ πρὸς ΕΓ, ἡ ΖΗ πρὸς ΗΓ [καὶ ἐναλλὰξ καὶ συνθέντι]. ὡς ἄρα συναμφότερος ἡ ΖΕ, ΕΓ πρὸς ΖΓ, ἡ ΕΓ πρὸς ΓΗ. ὥστε ἡ ΓΕ πρὸς ΓΗ μείζονα λόγον ἔχει, ἤπερ φοά πρὸς ρνγ′. ἡ ΕΗ ἄρα πρὸς ΗΓ δυνάμει λόγον ἔχει, ὃν $M^{\lambda\delta}$ ͵θυν′ πρὸς M^{β} ͵γυθ′. μήκει ἄρα, ὃν φμά η′′ πρὸς ρνγ′. πάλιν δίχα ἡ ὑπὸ



ΗΕΓ τῇ ΕΘ. διὰ τὰ αὐτὰ ἄρα ἡ ΕΓ πρὸς ΓΘ μείζονα λόγον ἔχει, ἢ ὃν ͵αρξβ′ η′′ πρὸς ρνγ′. ἡ ΘΕ ἄρα πρὸς ΘΓ μείζονα λόγον ἔχει, ἢ ὃν ͵αροβ′ η′′ πρὸς ρνγ′. ἔτι δίχα ἡ ὑπὸ ΘΕΓ τῇ ΕΚ. ἡ ΕΓ ἄρα πρὸς ΓΚ μείζονα λόγον ἔχει, ἢ ὃν ͵βτλδ′ δ′′ πρὸς ρνγ′. ἡ ΕΚ ἄρα πρὸς ΓΚ μείζονα, ἢ ὃν ͵βτλθ′ δ′′ πρὸς

2. *τρίτον*] τριτον (-τον per comp.) F, corr. B*. 3. *μείζονα λόγον* Wallis. *ὃν*] scripsi cum Eutocio; η ον F, uulgo.

Heiberg 1880 / printed p.265 / Latin / PDF page 284

sit circulus, et diametrus $A\Gamma$, et centrum E, et $\Gamma\Lambda Z$ linea circulum contingens, et $\angle ZE\Gamma$ tertia pars recti. itaque $EZ : Z\Gamma = 306 : 153$ [u. Eutocius], sed

$$E\Gamma : \Gamma Z = 265 : 153 \text{ [u. Eutocius].}$$

iam secetur $\angle ZE\Gamma$ in duas partes aequales linea EH. est igitur

$$ZE : E\Gamma = ZH : H\Gamma \text{ [Eucl. VI, 3].}$$

quare

$$ZE + E\Gamma : Z\Gamma = E\Gamma : \Gamma H \text{ [u. Eutocius].}^{1})$$

quare

$$\Gamma E : \Gamma H > 571 : 153 \text{ [u. Eutocius].}^{2})$$

itaque

$$EH^2 : H\Gamma^2 = 349450 : 23409 \text{ [u. Eutocius].}$$

itaque $EH : H\Gamma = 591\frac{1}{8} : 153$. rursus secetur eodem modo $\angle HE\Gamma$ linea $E\Theta$. propter eadem igitur erit

$$E\Gamma : \Gamma\Theta > 1162\tfrac{1}{8} : 153 \text{ [u. Eutocius].}$$

quare $\Theta E : \Theta\Gamma > 1172\frac{1}{8} : 153$ [u. Eutocius]. rursus secetur $\angle \Theta E\Gamma$ linea EK. erit

$$E\Gamma : \Gamma K > 2334\tfrac{1}{4} : 153 \text{ [u. Eutocius].}$$

---

1) Sequentia uerba lin. 6—7: *καὶ ἐναλλὰξ καὶ συνθέντι* a transscriptore ex Eutocio huc prauo ordine illata sunt.

2) Quae Archimedes breuissime, omissis computationibus, proponit, copiose et perspicue explicat Eutocius; quare satis habui lectorem ad eum reuocare. quo modo Archimedes numeros 153 et 780 inuenerit, aut quibus adiumentis instructus latera numerorum non quadratorum computauerit, nondum constat (Quaest. Arch. p. 60—66). haec propositio difficillima a transscriptore et fortasse etiam a librariis pessime habita est. citatur ab Archimede ipso Arenar. I, 19; II, 3 et a Simplicio in Aristot. IV p. 508, b.

---

7. *συνθέντι καὶ ἐναλλάξ* Wallis. 10. *μείζονα λόγον* Wallis. *ἢ ὃν* Wallis. idem post *ἄρα* lin. 11 addit *μείζονα ἤ*. 17. *μείζονα*] scripsi; *μειζον* F, uulgo; *μείζονα λόγον ἔχει* Wallis.

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Heiberg 1880 / printed p.266 / Greek / PDF page 285

ρνγ΄. ἔτι δίχα ἡ ὑπὸ ΚΕΓ τῇ ΛΕ. ἡ ΕΓ ἄρα πρὸς ΛΓ μείζονα [μήκει] λόγον ἔχει, ἥπερ ,δχογ΄ ʟ″ πρὸς ρνγ΄. ἐπεὶ οὖν ἡ ὑπὸ ΖΕΓ τρίτον οὖσα ὀρθῆς τέτμηται τετράκις δίχα, ἡ ὑπὸ ΛΕΓ ὀρθῆς ἐστι μη″. κείσθω οὖν αὐτῇ ἴση πρὸς τῷ Ε ἡ ὑπὸ ΓΕΜ. ἡ ἄρα ὑπὸ ΛΕΜ ὀρθῆς ἐστι κδ″. καὶ ἡ ΛΜ ἄρα εὐθεῖα τοῦ περὶ τὸν κύκλον ἐστὶ πολυγώνου πλευρὰ πλευρὰς ἔχοντος ϙ΄. ἐπεὶ οὖν ἡ ΕΓ πρὸς τὴν ΓΛ ἐδείχθη μείζονα λόγον ἔχουσα, ἥπερ ,δχογ΄ ʟ″ πρὸς ρνγ΄, ἀλλὰ τῆς μὲν ΕΓ διπλῆ ἡ ΑΓ, τῆς δὲ ΓΛ διπλασίων ἡ ΛΜ, καὶ ἡ ΑΓ ἄρα πρὸς τὴν τοῦ ϙ΄ πολυγώνου περίμετρον μείζονα λόγον ἔχει, ἥπερ ,δχογ΄ ʟ″ πρὸς α
Μ ,δχπη΄. καί ἐστιν τριπλασία, καὶ ὑπερέχουσιν χξζ΄ ʟ″, ἅπερ τῶν ,δχογ΄ ʟ″ ἐλάττονά ἐστιν ἢ τὸ ἕβδομον. ὥστε τὸ πολύγωνον τὸ περὶ τὸν κύκλον τῆς διαμέτρου ἐστὶ τριπλάσιον καὶ ἐλάττονι ἢ τῷ ἑβδόμῳ μέρει μεῖζον. ἡ τοῦ κύκλου ἄρα περίμετρος πολὺ μᾶλλον ἐλάσσων ἐστὶν ἢ τριπλασίων καὶ ἑβδόμῳ μέρει μείζων.

ἔστω κύκλος, καὶ διάμετρος ἡ ΑΓ, ἡ δὲ ὑπὸ ΒΑΓ τρίτον ὀρθῆς. ἡ ΑΒ ἄρα πρὸς ΒΓ ἐλάσσονα λόγον ἔχει, ἢ ὃν ,ατνά πρὸς ψπ΄ [ἡ δὲ ΑΓ πρὸς ΓΒ, ὃν ,αφξ΄ πρὸς ψπ΄].

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2. μήκει delet Wallis; om. Eutocius. ,δχογ΄ ʟ″] ,δνογ FV. 5. ἴση η F; corr. Wallis. idem post ΓΕΜ addit: καὶ ἐκβεβλήσθω ἡ ΖΓ ἐπὶ τὸ Μ. 6. post εὐθεῖα ed. Basil. addit πλευρά ἐστιν (ἐστι Wallis), omisso ἐστι lin. 7, quod habent F (per comp.), cett. codd. 7. ante πολυγώνου ed. Basil. habet περιγραφομένου. πλευρά] addidit Wurm; om. F, uulgo. 11. post ΛΜ addit Wallis: καὶ ἡ ΑΓ ἄρα πρὸς τὴν ΛΜ μείζονα λόγον ἔχει, ἥπερ ,δχογ΄ ʟ″ πρὸς ρνγ΄. 13. ante καί idem: ἀνάπαλιν ἄρα ἡ περίμετρος τοῦ πολυγώνου πρὸς τὴν διάμετρον ἐλάσσονα λόγον ἔχει, ἥπερ α
Μ ,δχπη΄ πρὸς ,δχογ΄ ʟ″. 14. ἤ]

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Heiberg 1880 / printed p.267 / Latin / PDF page 286

quare $EK : \Gamma K > 2339\frac{1}{4} : 153$ [u. Eutocius]. rursus secetur $\angle KE\Gamma$ linea ΔE. erit igitur

$$E\Gamma : \Delta\Gamma > 4673\frac{1}{2} : 153 \text{ [u. Eutocius].}$$

iam quoniam $\angle ZE\Gamma$, qui tertia pars est recti, quater in partes aequales diuisus est, $\angle \Delta E\Gamma$ erit pars duodequinquagesima recti. ponatur[1]) igitur ei aequalis $\angle \Gamma EM$ ad punctum E. itaque $\angle \Delta EM$ pars uicesima quarta est recti. quare linea ΔM latus est polygoni 96 latera habentis circum circulum circumscripti. et quoniam demonstratum est $E\Gamma : \Gamma\Delta > 4673\frac{1}{2} : 153$, et $\Delta\Gamma = 2E\Gamma$, $\Delta M = 2\Gamma\Delta$, $\Delta\Gamma$ etiam ad perimetrum polygoni 96 latera habentis maiorem habet rationem, quam $4673\frac{1}{2} : 14688$ [u. Eutocius]. est igitur triplo maior [perimetrus polygoni], et supersunt $667\frac{1}{2}$, quod minus est septima parte $4673\frac{1}{2}$. itaque [perimetrus] polygoni circumscripti minor est quam triplo et septima parte maior diametro. quare ambitus circuli multo magis[2]) minor est quam triplo et septima parte maior diametro.

sit circulus, et diametrus $\Delta\Gamma$, et $\angle BA\Gamma$ tertia pars recti. itaque $AB : B\Gamma < 1351 : 780$ [u. Eutocius].

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1) Quamquam Eutocius: *κείσθω οὖν, φησι, ἴση αὐτῇ ἡ ὑπὸ ΓΕΜ*, tamen ex sequentibus adparet, eum suis ipsius uerbis uti. quare ne infra quidem (lin. 8: *δέδεικται*, lin. 9: *ρνγ΄, καὶ ἔστι τῆς*) constat, eum genuinam formam praebere. sed lin. 19—20 puto eum recte praebere: *κύκλος περὶ διάμετρον τὴν ΑΓ καὶ τρίτον ὀρθῆς ἡ ὑπὸ ΒΑΓ*; lin. 10 om. *διπλασίων*. de lin. 10, 11, 15, 21 u. p. 269 not. 1.

2) Perimetrus enim polygoni maior est ambitu circuli; de sph. et cyl. I, 1.

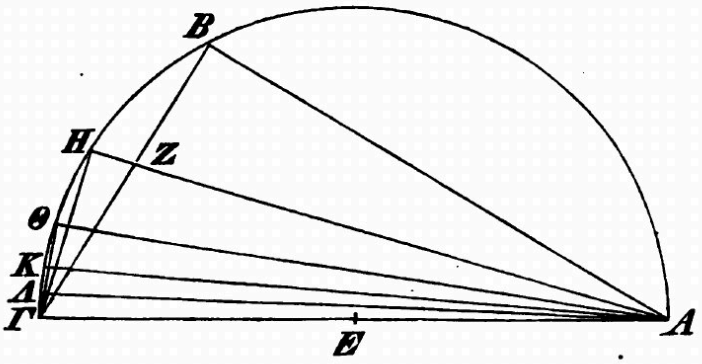
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om. F; corr. Wallis. 16. *ἐλάττονι*] scripsi; *ελαττον* F, uulgo. 19. *Δ΄* addit F; corr. Wallis. 20. *τριτον* F; corr. B*. 21. *΄ατνα΄*] *τνα* F; corr. B manu 2.*

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Heiberg 1880 / printed p.268 / Greek / PDF page 287

δίχα ἡ ὑπὸ *BAΓ* τῇ *AH*. ἐπεὶ ἴση ἐστὶν ἡ ὑπὸ *BAH* τῇ ὑπὸ *HΓB*, ἀλλὰ καὶ τῇ ὑπὸ *HAΓ*, καὶ ἡ ὑπὸ *HΓB* τῇ ὑπὸ *HAΓ* ἐστιν ἴση. καὶ κοινὴ ἡ ὑπὸ *AHΓ* ὀρθή. καὶ τρίτη ἄρα ἡ ὑπὸ *HZΓ* τρίτῃ τῇ ὑπὸ *AΓH* ἴση. ἰσογώνιον ἄρα τὸ *AHΓ* τῷ *ΓHZ*



τριγώνῳ. ἔστιν ἄρα, ὡς ἡ *AH* πρὸς *HΓ*, ἡ *ΓH* προς *HZ*, καὶ ἡ *AΓ* πρὸς *ΓZ*. ἀλλ' ὡς ἡ *AΓ* πρὸς *ΓZ*, καὶ συναμφότερος ἡ *ΓAB* πρὸς *BΓ*. καὶ ὡς συναμφότερος ἄρα ἡ *BAΓ* πρὸς *BΓ*, ἡ *AH* πρὸς *HΓ*. διὰ τοῦτο οὖν ἡ *AH* πρὸς τὴν *HΓ* ἐλάσσονα λόγον ἔχει, ἤπερ βϡιά πρὸς ψπ΄, ἡ δὲ *AΓ* πρὸς τὴν *ΓH* ἐλάσσονα, ἢ ὃν ,γιγ΄ ι΄΄ δ΄΄ πρὸς ψπ΄. δίχα ἡ ὑπὸ *ΓAH* τῇ *AΘ*. ἡ *AΘ* ἄρα διὰ τὰ αὐτὰ πρὸς τὴν *ΘΓ* ἐλάσσονα λόγον ἔχει, ἢ ὃν ,εϡκδ΄ ι΄΄ δ΄΄ πρὸς ψπ΄, ἢ ὃν ,αωκγ΄ πρὸς σμ΄. ἑκατέρα γὰρ ἑκατέρας δ΄ ιγ΄΄. ὥστε ἡ *AΓ* πρὸς τὴν *ΓΘ*, ἢ ὃν ,αωλη΄ ϑ΄ ια΄΄ πρὸς σμ΄. ἔτι δίχα ἡ ὑπὸ *ΘAΓ* τῇ *KA*. καὶ ἡ *AK* πρὸς τὴν *KΓ* ἐλάσ-

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1. Ante δίχα ed. Basil. habet τετμήσθω. 3. τῇ] ἄρα τῇ ed. Basil. 4. ἄρα] scripsi; εσται F, uulgo; ἄρα ἴση ἔσται ed. Basil., Torellius. 5. ἴση] addidi; om. F, uulgo. 8. *ΓA*, *AB* Torellius. 9. *BA*, *AΓ* Nizze. *AH*] *ΔH* F; corr. B mg.* 12. pro ι΄΄ FBC* habent *Γ*΄. 14. ,εϡκδ΄ ι΄΄] ετκδ ε΄ F; corr. ed. Basil. (λ pro ϡ; corr. Wallis). 15. σμ΄] σν F; corr. ed. Ba-

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Heiberg 1880 / printed p.269 / Latin / PDF page 288

secetur[1]) $\angle BA\Gamma$ in partes aequales linea AH. iam quoniam $\angle BAH = H\Gamma B$ [Eucl. III, 26], sed etiam $= HA\Gamma$, erit $H\Gamma B = HA\Gamma$. et communis est $\angle AH\Gamma$ rectus [Eucl. III, 31]. quare etiam $HZ\Gamma = A\Gamma H$ [Eucl. I, 32]. quare triangula $AH\Gamma$, ΓHZ angulos aequales habent. est igitur [Eucl. VI, 4]

$$AH : H\Gamma = \Gamma H : HZ = A\Gamma : \Gamma Z.$$

sed $A\Gamma : \Gamma Z = \Gamma A + AB : B\Gamma$ [Eucl. VI, 3; Eutocius]. quare $\Gamma A + AB : B\Gamma = AH : H\Gamma$. itaque $AH : H\Gamma < 2911 : 780$ [u. Eutocius],[2]) et

$$A\Gamma : \Gamma H < 3013\tfrac{1}{2}\tfrac{1}{4} : 780 \text{ [u. Eutocius].}$$

secetur eodem modo $\angle \Gamma AH$ linea $A\Theta$. propter eadem igitur erit $A\Theta : \Theta\Gamma < 5924\tfrac{1}{2}\tfrac{1}{4} : 780$ [u. Eutocius], hoc est $< 1823 : 240$. altera[3]) enim alterius $\frac{4}{13}$ [u. Eutocius]. quare est $A\Gamma : \Gamma\Theta < 1838\frac{9}{11} : 240$ [u. Eutocius]. porro secetur $\angle \Theta A\Gamma$ linea KA. est igitur

1) Cum p. 266, 20—21; 268, 9—12; 13—16; 268, 17—270, 1 ab Eutocio non ipsis uerbis Archimedis citari uideantur, has contra scripturas in lemmatis eius seruatas genuinas putauerim et in uerbis Archimedis a transscriptore mutatas: lin. 1: τετμήσθω δίχα; ἐπεὶ οὖν; lin. 3: ἄρα τῇ; lin. 4: λοιπή et λοιπῇ pro τρίτη et τρίτῃ; lin. 5: ἔστιν ἴση; ἄρα ἐστί; ΑΗΓ τρίγωνον; lin. 8 καί (prius) om.; lin. 16: πρὸς ΘΓ ἐλάσσονα λόγον ἔχει ἤπερ; lin. 15: ἐστὶ δ' ιγ''; lin. 17: ΘΑΓ γωνία. simul alia transscriptionis uestigia colligam: ut lin. 5 om. τρίγωνον prop. 1 p. 260, 14; 2 p. 262, 6; διπλῆ p. 266, 10 (διπλασίων Nizze; cfr. prop. 2 p. 262, 5); τοῦ ϛ πολυγώνου p. 266, 11; 270, 9; τὸ πολυγώνον pro ἡ περίμετρος τοῦ πολυγώνου p. 266, 15 (ἡ περίμετρος τοῦ πολυγώνου τοῦ — τριπλασίων — μείζων Nizze). praeterea Eutocius uerba ἡ δὲ ΑΓ — ψπ' p. 266, 21 habuisse non uidetur; debebat insuper esse ἡ γὰρ ΑΓ.

2) Hic, ut saepissime in hac propositione, utitur proportione illa, quam exposui Quaest. Arch. p. 48.

3) Genus femininum refertur ad auditum uerbum πλευρά.

sil.* ἑκατέρας] ἑκατέρων Wallis. ιγ''] ιγ' α' F; corr. ed. Basil. 16. Post ΓΘ additur ἐλάσσονα λόγον ἔχει in ed. Basil. ια''] om. F; corr. Wallis.

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Heiberg 1880 / printed p.270 / Greek / PDF page 289

σονα ἄρα λόγον ἔχει, ἢ ὃν ,αξ′ πρὸς ξϛ′. ἑκατέρα γὰρ ἑκατέρας ια′ μ″. ἡ *ΑΓ* ἄρα πρὸς τὴν *ΓΚ*, ἢ ὃν ,αθ′ ϛ″ πρὸς ξϛ′. ἔτι δίχα ἡ ὑπὸ *ΚΑΓ* τῇ *ΛΑ*. ἡ *ΑΛ* ἄρα πρὸς τὴν *ΛΓ* ἐλάσσονα λόγον ἔχει, ἢ ὃν τὰ ,βιϛ′ ϛ″ πρὸς ξϛ′, ἡ δὲ *ΑΓ* πρὸς *ΓΛ* ἐλάσσονα, ἢ τὰ ,βιζ′ δ″ πρὸς ξϛ′. ἀνάπαλιν ἄρα ἡ περίμετρος τοῦ πολυγώνου πρὸς τὴν διάμετρον μείζονα λόγον ἔχει, ἤπερ ,στλϛ′ πρὸς ,βιζ′ δ″, ἅπερ τῶν ,βιζ′ δ″ μείζονά ἐστιν ἢ τριπλασίονα καὶ δέκα ὁα″. καὶ ἡ περίμετρος ἄρα τοῦ ϥϛ′ πολυγώνου τοῦ ἐν τῷ κύκλῳ τῆς διαμέτρου τριπλασίων ἐστὶ καὶ μείζων ἢ ι′ οα″. ὥστε καὶ ὁ κύκλος ἔτι μᾶλλον τριπλασίων ἐστὶ καὶ μείζων ἢ ι′ οα″.

ἡ ἄρα τοῦ κύκλου περίμετρος τῆς διαμέτρου τριπλασίων ἐστὶ καὶ ἐλάσσονι μὲν ἢ ἑβδόμῳ μέρει, μείζονι δὲ ἢ ι′ οα″ μείζων.

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1. Post ἢ ὃν addit Wallis: ,γχξα′ θ′ ια″ πρὸς σμ′ ἢ ὃν. ξϛ′] cξϛ F; corr. ed. Basil. 2. ἑκατέρας] ed. Basil. ex Eutocio; εκατερα FBC*; ἑκατέρων Wallis. ια′ μ″ ἡ *ΑΓ*] οιμαι F; corr. Wallis. *ΓΚ* ἢ ὃν] scripsi cum Wurmio; καταγον F; κατάλογον ed. Basil.; *ΓΚ* ἐλάσσονα λόγον Wallis. ,αθ′ ϛ″] scripsi; ,αος F, uulgo; ἔχει ἢ ,αθ′ ϛ′ Wallis. 4. *ΛΓ*] *ΑΓ* F; corr. Wallis. 6. Post ἄρα ἡ Wallis addit: *ΑΓ* πρὸς τὴν *ΓΛ* μείζονα λόγον ἔχει ἤπερ ξϛ′ πρὸς βιζ′ δ′· καί (ἡ addit Nizze). 7. ,στλϛ′] ,στας F; corr. Wallis. 8. ,βιζ′] (prius) ,ιξ F; corr. Wallis. 9. οα″] ο′ α′ F; corr. Wallis. 11. ι′ οα″] scripsi; ὃν ο′ ια′ F, uulgo; δέκα οα′ ed. Basil. Tor., Wall. 13. ι′ οα″] scripsi; θ′ ια′ F, uulgo; δέκα οα′ ed. Basil. Tor., Wall. 14. ἐλάσσονι] scripsi; ελασσων F, uulgo. μείζονι δὲ ἢ ι′ οα″ μείζων] scripsi; μειζων δε F, uulgo; μείζων δὲ ἢ δέκα ἑβδομηκοστομόνοις ὑπερέχουσα Wallis.

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Heiberg 1880 / printed p.271 / Latin / PDF page 290

$AK : K\Gamma < 1007 : 66$ [u. Eutocius]. altera enim alterius est $\frac{1}{4}\frac{1}{0}$. itaque.

$$A\Gamma : \Gamma K < 1009\tfrac{1}{6} : 66 \text{ [u. Eutocius].}$$

porro secetur $\angle K A \Gamma$[1]) linea ΛA. erit igitur

$$A\Lambda : \Lambda\Gamma < 2016\tfrac{1}{6} : 66 \text{ [u. Eutocius],}$$

et $A\Gamma : \Gamma\Lambda < 2017\frac{1}{4} : 66$ [u. Eutocius]. et e contrario [$\Gamma\Lambda : A\Gamma > 66 : 2017\frac{1}{4}$ (Pappus VII, 49 p. 688); sed $\Gamma\Lambda$ latus est polygoni 96 latera habentis. quare][2]) perimetrus polygoni ad diametrum maiorem rationem habet quam $6336 : 2017\frac{1}{4}$, quod maius est quam triplo et $\frac{1}{7}\frac{0}{1}$ maius quam $2017\frac{1}{4}$. itaque perimetrus polygoni inscripti 96 latera habentis[3]) maior est quam triplo et $\frac{1}{7}\frac{0}{1}$ maior diametro. quare etiam multo magis[4]) circulus maior est quam triplo et $\frac{1}{7}\frac{0}{1}$ maior diametro. itaque ambitus circuli triplo maior est diametro et excedit spatio minore quam $\frac{1}{7}$, maiore autem quam $\frac{1}{7}\frac{0}{1}$.[5])

1) $KA\Gamma$ *γωνία* lin. 3 Eutocius. ceteras huius paginae discrepantias, quae apud eum inueniuntur, inde ortas esse puto, quod Archimedis demonstrationem non ad uerbum citauit, sed suis uerbis reddidit.

2) Ueri simile est, Archimedem ipsum haec addidisse.

3) *τοῦ ϛ′ πολυγώνου* transscriptori debetur, sicut etiam lin. 11: *ὁ κύκλος* pro *ἡ τοῦ κύκλου περίμετρος* (*περιφέρεια*).

4) Quippe quae maior est perimetro polygoni (de sph. et cyl. I p. 10).

5) *Ἀρχιμήδους κυκλου μετρησις* in fine F, Cr.

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